

23CSE111 OBJECT ORIENTED PROGRAMMING

S2 B.Tech CSE

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

Group No :D2

Team Members :

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1. Problem Title :

Online Food Delivery Management System

2. Problem Description :

Customers want convenient food ordering. Phone-based systems cause delays and errors. A centralized digital platform is needed. Modern lifestyles have increased demand for convenient food ordering. Customers prefer ordering through apps instead of visiting restaurants. However, phone-based ordering leads to miscommunication, delays, and incorrect deliveries. Restaurants struggle to manage multiple orders efficiently. Delivery coordination is another challenge. Without a structured system, delays and dissatisfaction occur. An Online Food Delivery System integrates customers, restaurants, and delivery agents on a single platform, ensuring smooth order processing and tracking.

3. Identified Entities (Classes)

Class Name	Description
User	Records personal details of customer like name, address.
Order	It stores information about ordered items, total cost, order status, and delivery time.
Restaurants	Stores restaurant information such as name, address, rating.

Delivery	It contains details like agent ID, contact information, and current location for tracking.
Menu	It contains details such as item name, price, category, and availability status.

4. Class Attributes and Behaviours

Class Name	Attributes	Methods(behaviours)
User	Username, DeliveryAddress, PhoneNumber, Feedback, Email	Login() Feedback() PaymentType() setOrder() setRestaurant() checkUser()
Order	OrderID OrderStatus OrderedItems Amount	CalculateAmount() PlaceOrder() CancelOrder() ConfirmOrder() saveOrder() getOrderID()

Restaurants	Restaurants Location Rating MenuItems distance	ReceiveOrder() AddMenuItem() BrowseRestaurants()
Menu	ItemName Price Category Availability	DisplayItem() UpdateAvailability() UpdatePrice()
Delivery	AgentName MobileNumber CurrentLocation VehicleNumber AvailabilityStatus	AcceptDelivery() UpdateLocation() MarkDelivered() TrackOrder() MakePayment()